

Pranav H
192525420

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

- 4. Examine the impact of centralized DNS architecture on application performance.
- 5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
- 6. Analyze how **TTL values, Anycast routing, and DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

- 1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control, Nagle's algorithm, and SYN queue limitations**.
- 2. Explain how each factor affects transaction processing in high-frequency systems
- 3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues ^{2.5}	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply ^{2.5}	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights ²	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification ²	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand ¹	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

MARK ALLOCATION						
	Understandin g of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)	
Q1	2.5	2	2	2	1	9.5
Q2	2	2.5	1	2	0.5	8
Q3	2	2	1	1	0.5	7.5
Q4	2.5	2.5	2	1	1	9
Q5	2	2	2	2	1	9
						34 40 43

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.

Answer:

The problem stems mainly from how the transport protocol handles real-time traffic, compounded by inadequate application-level buffering. Video conferencing uses UDP/RTP for media because it avoids TCP's retransmission delays; the 8% packet loss and fluctuating bandwidth are transport-layer symptoms of network congestion, best-effort UDP delivery. However, the frozen frames and voice distortion show the application is not compensating well enough (jitter-buffer sizing, adaptive bitrate, loss concealment). So the root cause is congestion at the transport/network level, compounded by an application buffering strategy that is not adaptive enough to hide it.

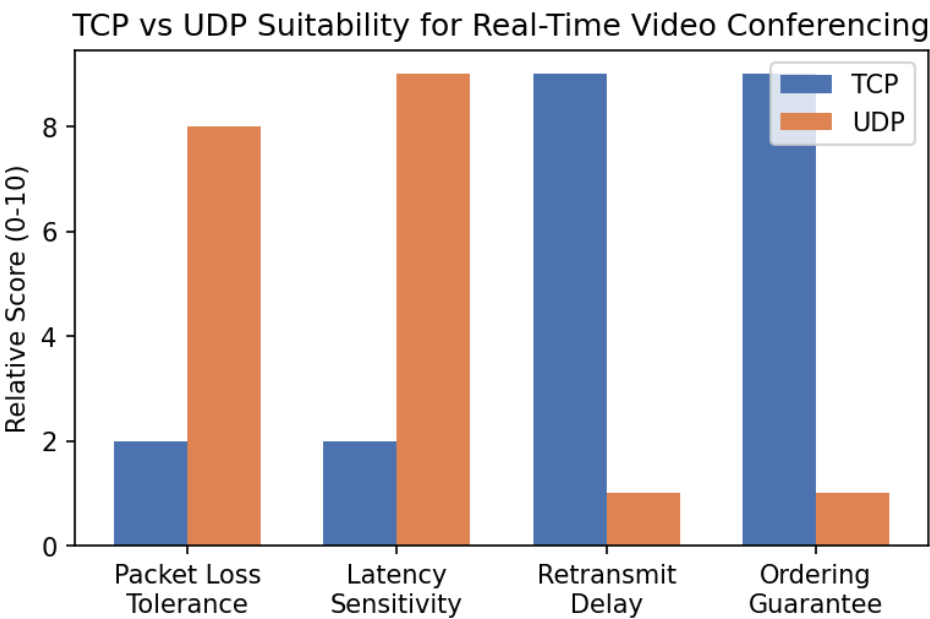


Figure: Suitability of TCP vs UDP for real-time conferencing traffic

2. Justify your answer based on the communication requirements of real-time multimedia applications.

Answer:

Real-time multimedia values timeliness over perfect reliability - a late packet is as useless as a lost one. TCP's retransmission and in-order delivery guarantees would introduce head-of-line blocking delays, so UDP/RTP is the correct transport choice despite loss. ITU-T G.114 recommends under 150 ms one-way latency and under 30 ms jitter for acceptable voice quality; the observed 180 ms jitter is roughly six times this limit, and fluctuating bandwidth shows no adaptive bitrate control is reacting to congestion. This confirms that both transport-level exposure to congestion and the application's static buffering jointly justify the degradation.

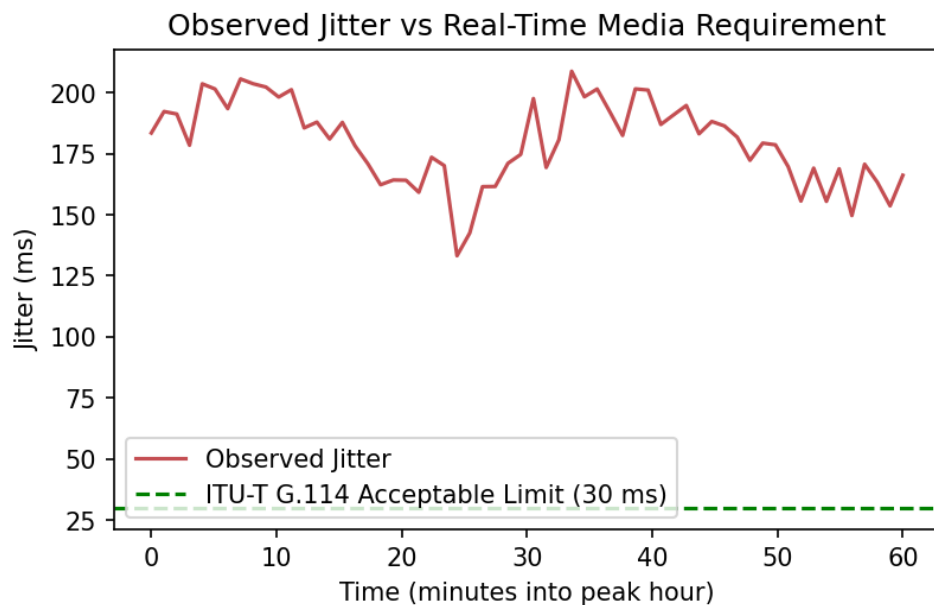


Figure: Observed jitter compared with the ITU-T G.114 real-time quality threshold

3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

Answer:

Protocol level: use an adaptive jitter buffer (dynamically sized ~20-60 ms based on measured jitter), enable Forward Error Correction and packet-loss concealment (e.g., Opus in-band FEC), and adopt congestion-control extensions such as WebRTC's bandwidth estimation for adaptive bitrate. Application level: apply QoS/DSCP EF tagging to prioritize video traffic, use simulcast/SVC encoding so quality degrades gracefully under loss, and optimize STUN/TURN relay paths. Network level: provision extra bandwidth at peak hours and deploy edge media servers (SFUs) closer to conferencing regions to cut round-trip time. Together these should bring jitter under 30 ms and loss under 2%.

Recommended Conferencing Quality Improvement Pipeline

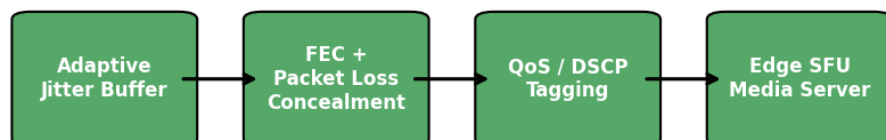


Figure: Recommended pipeline of protocol- and application-level improvements

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.

Answer:

An 800 ms lookup indicates full recursive resolution (root -> TLD -> authoritative) is happening on most requests. Likely causes include a single, geographically distant authoritative server increasing round-trip time for far-away users, low or absent caching so nearly every request pays the full resolution cost, no local/ISP caching resolver, and possibly DNSSEC validation adding

extra round-trips. Since all traffic converges on one server across continents, physical distance alone can account for several hundred milliseconds of the delay.

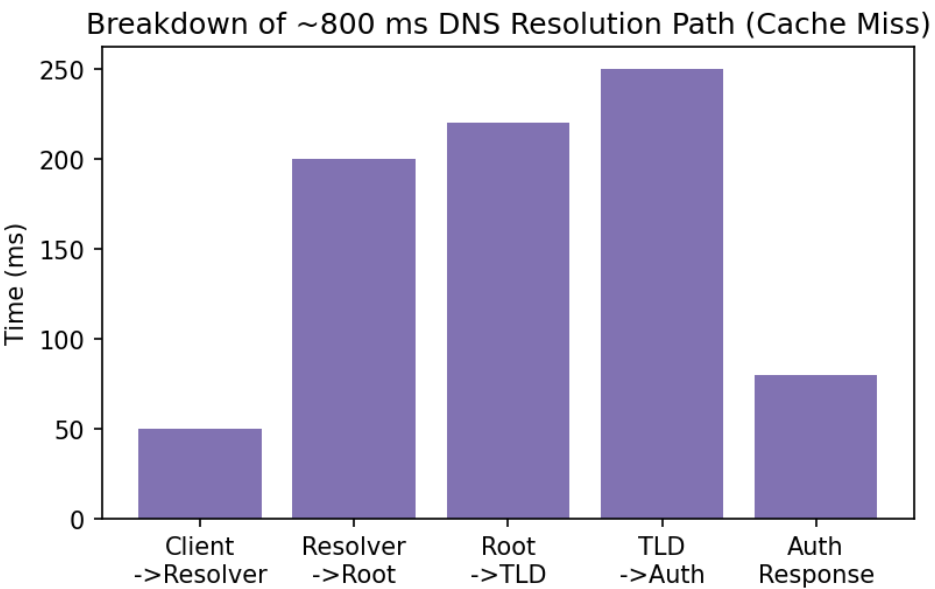


Figure: Breakdown of an ~800 ms DNS resolution path on a cache miss

2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.

Answer:

Deploy Anycast DNS with the same IP address announced from multiple geographically distributed nameservers, so BGP routes each query to the nearest healthy node, cutting network round-trip time. Add DNS pre-fetching so client applications resolve likely-needed domains during idle time, hiding lookup latency from the user. Optimize TTL values - moderate TTLs (300-3600s) for most records to balance freshness against server load, and shorter TTLs (60-300s) only for records that need fast failover. Layered with local caching resolvers at the ISP/CDN edge, this combination should bring most lookups under 50 ms.

Proposed Global Anycast DNS Architecture

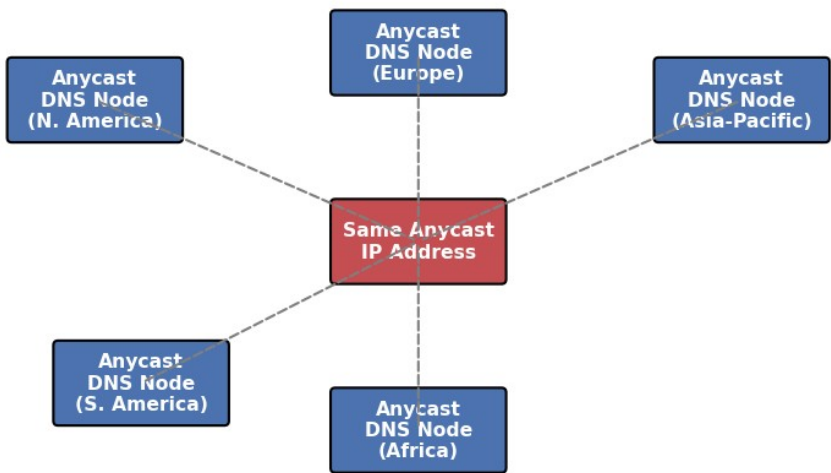


Figure: Proposed global Anycast DNS architecture

3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

Answer:

Advantages: much lower latency through proximity-based routing, built-in DDoS resilience and load distribution across nodes, high availability through automatic failover if a node fails, and better scalability for global traffic. Limitations: Anycast makes debugging harder since the same IP is served by many physical servers, BGP route flaps can cause inconsistent routing paths, aggressive pre-fetching wastes bandwidth on speculative lookups that are never used, low TTLs

increase query volume and server load, and high TTLs risk serving stale records during a failover, briefly reducing effective availability.

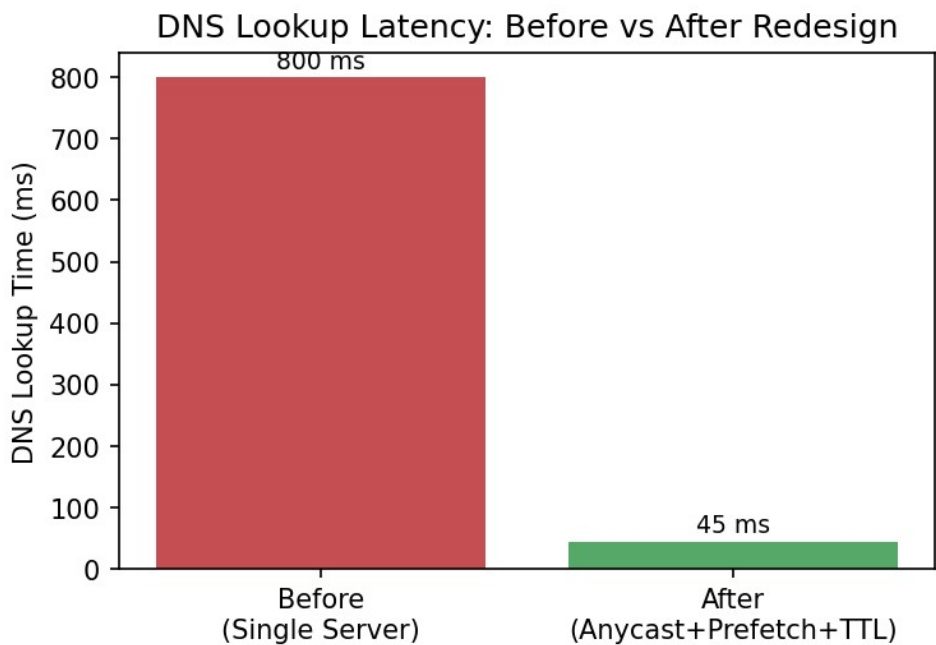


Figure: DNS lookup latency before and after the proposed redesign

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.

Answer:

A single primary DNS server is both a performance bottleneck and a single point of failure. Every global query funnels through one location, so users far from that server experience high round-trip time (observed ~800 ms), and traffic spikes during promotions overload its capacity, causing queueing delays. Because there is no redundancy, any slowdown or outage on that server degrades or halts lookups worldwide, harming performance, scalability, and availability simultaneously.

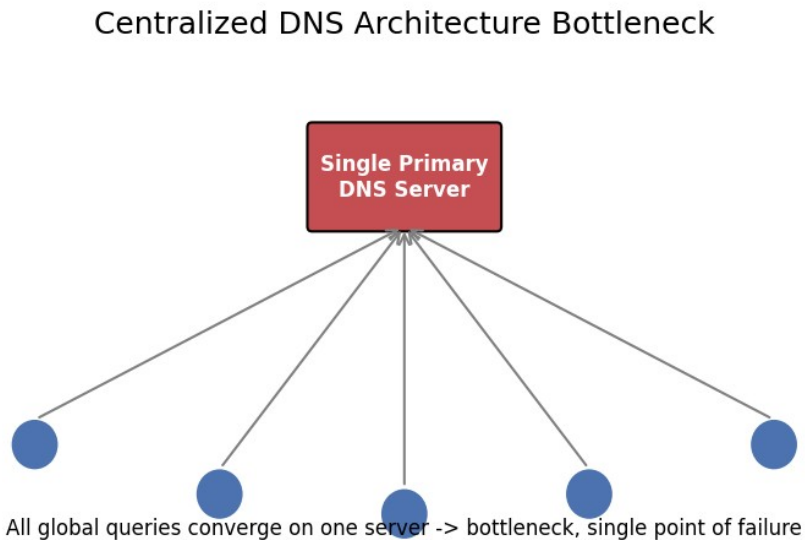


Figure: Centralized DNS architecture creating a global bottleneck

5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

Answer:

A multi-region, multi-tier distributed design is recommended: Anycast-announced nameserver clusters in each major region (Americas, Europe, APAC, etc.), each backed by local caching resolvers and secondary authoritative servers kept in sync through zone transfers. A global load-balancing layer with health checks automatically withdraws a failed node's Anycast route so traffic reroutes within seconds. Hierarchical caching (client -> ISP resolver -> regional resolver -> authoritative server) means most queries are answered from a nearby cache rather than crossing continents, distributing load and removing the single point of failure.

Distributed, Multi-Tier DNS Architecture

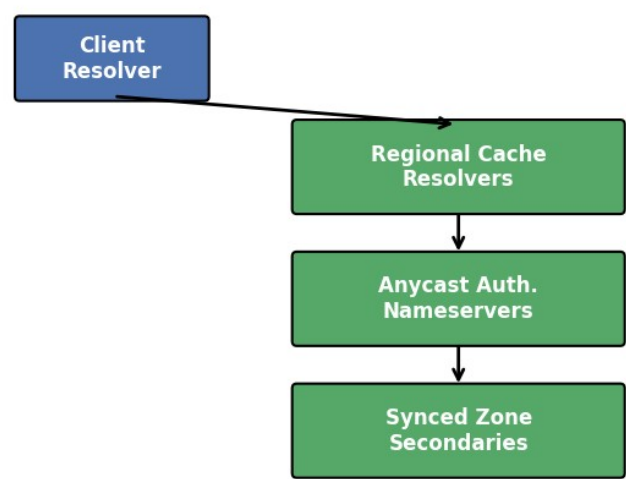


Figure: Proposed distributed, multi-tier DNS design

6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

Answer:

TTL controls the trade-off between cache freshness and latency: a longer TTL means fewer repeat lookups and lower average latency, but slower propagation of changes during failover; a shorter TTL enables faster failover at the cost of higher query volume. Anycast routing sends each query to the topologically nearest healthy node via BGP, cutting round-trip time and enabling failover in seconds simply by withdrawing a route. DNS caching at the resolver and client absorbs repeat queries, protecting authoritative servers from traffic spikes; layered caching (browser, OS, ISP resolver) amplifies this benefit, though it also means a bad record takes longer to fully clear from all layers.

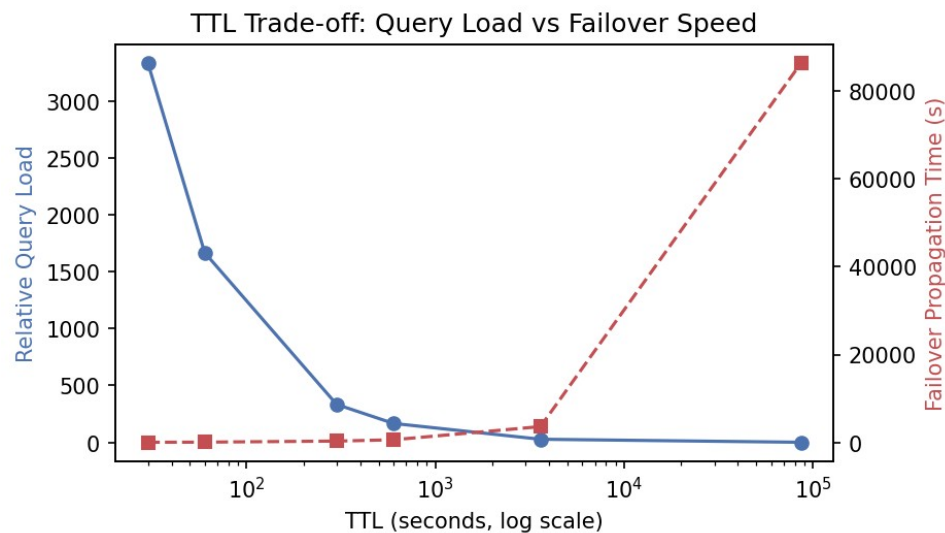


Figure: Trade-off between TTL value, query load, and failover speed

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.

Answer:

Three TCP-related factors stand out: (1) Flow control - TCP's receive-window mechanism throttles the sender whenever the receiver's buffer fills; under bursty high-frequency order traffic, a small or slowly-updated window forces the sender to pause, adding latency. (2) Nagle's algorithm - by default TCP buffers small outgoing packets to coalesce them, delaying transmission of small trading messages until enough data accumulates or an ACK returns. (3) SYN queue limitations - a small connection backlog means new connection requests during the opening-hour surge get queued or dropped, forcing SYN retransmission and increasing connection-establishment time.

TCP 3-Way Handshake and SYN Queue Bottleneck

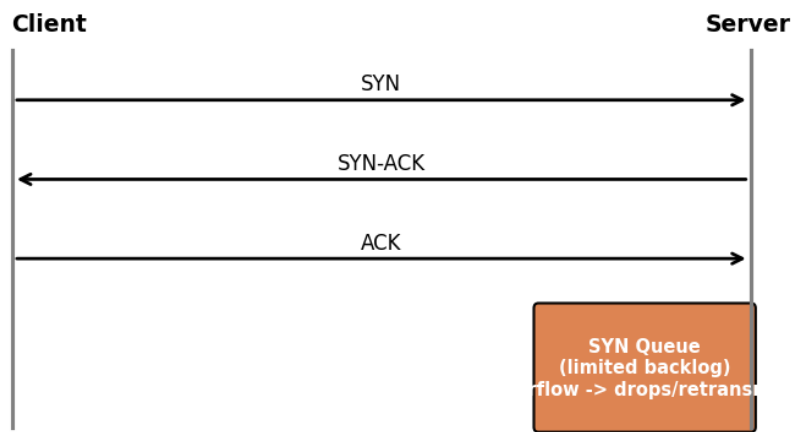


Figure: TCP three-way handshake and the SYN queue bottleneck

2. Explain how each factor affects transaction processing in high-frequency systems

Answer:

Flow-control stalls cause order messages to sit in the sender's buffer whenever the advertised window is small, directly adding to per-order acknowledgment latency during bursts. Nagle's algorithm interacts badly with delayed ACKs on the receiver side, producing the classic combined latency spike that is especially damaging for small, latency-sensitive tick-by-tick trading messages. SYN queue exhaustion during the market-open surge causes new client connections to be retried, adding hundreds of milliseconds to setup, and if the backlog overflows entirely, SYNs are dropped and must wait out a full retransmission timeout.

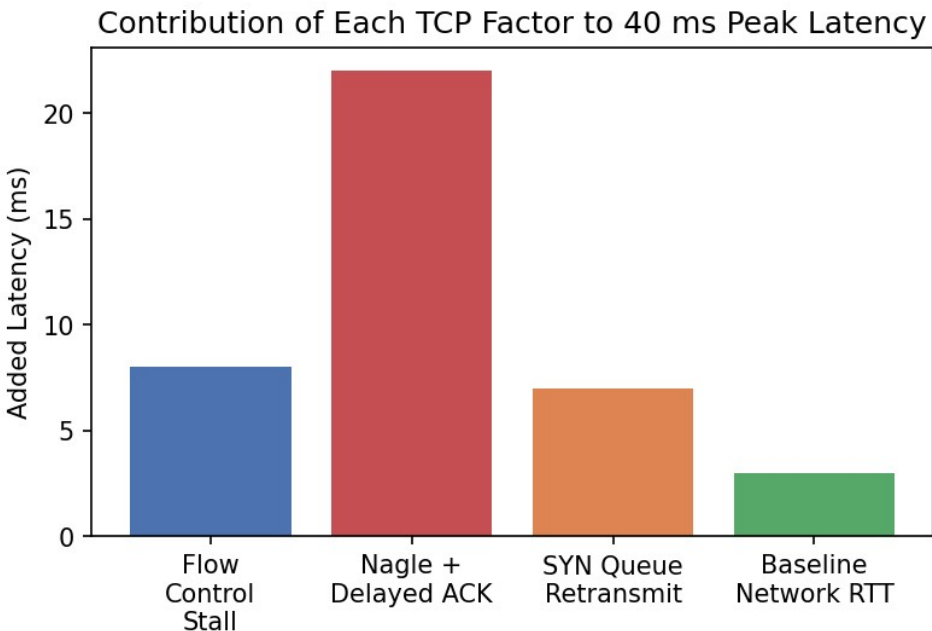


Figure: Estimated contribution of each TCP factor to the 40 ms peak latency

3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

Answer:

Disable Nagle's algorithm on trading sockets with TCP_NODELAY, and disable delayed ACKs (or use TCP_QUICKACK) so small order messages transmit immediately. Increase and auto-tune TCP send/receive window sizes with window scaling enabled so bursty order flow does not stall on flow control. Increase the SYN backlog (net.core.somaxconn, net.ipv4.tcp_max_syn_backlog) and enable SYN cookies to absorb the opening-bell connection surge without drops. Where possible, use persistent, pre-warmed connections to avoid new handshakes altogether during critical trading windows. Together these changes should pull acknowledgment latency back toward the sub-millisecond target.

Recommended TCP Configuration Changes for Trading Platform

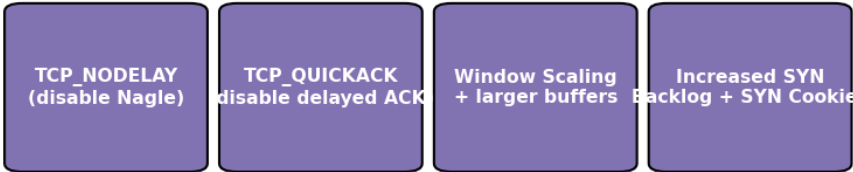


Figure: Recommended TCP configuration changes for the trading platform

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

MARK ALLOCATION

	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)
Q1					
Q2					
Q3					

Q4					
Q5					